CLMSynth · run pipeline

clmsynth <config> → YYYYMMDD_HHMMSS_Source/{csv, png, info} · one process per config

CLMSynth v0.7.5

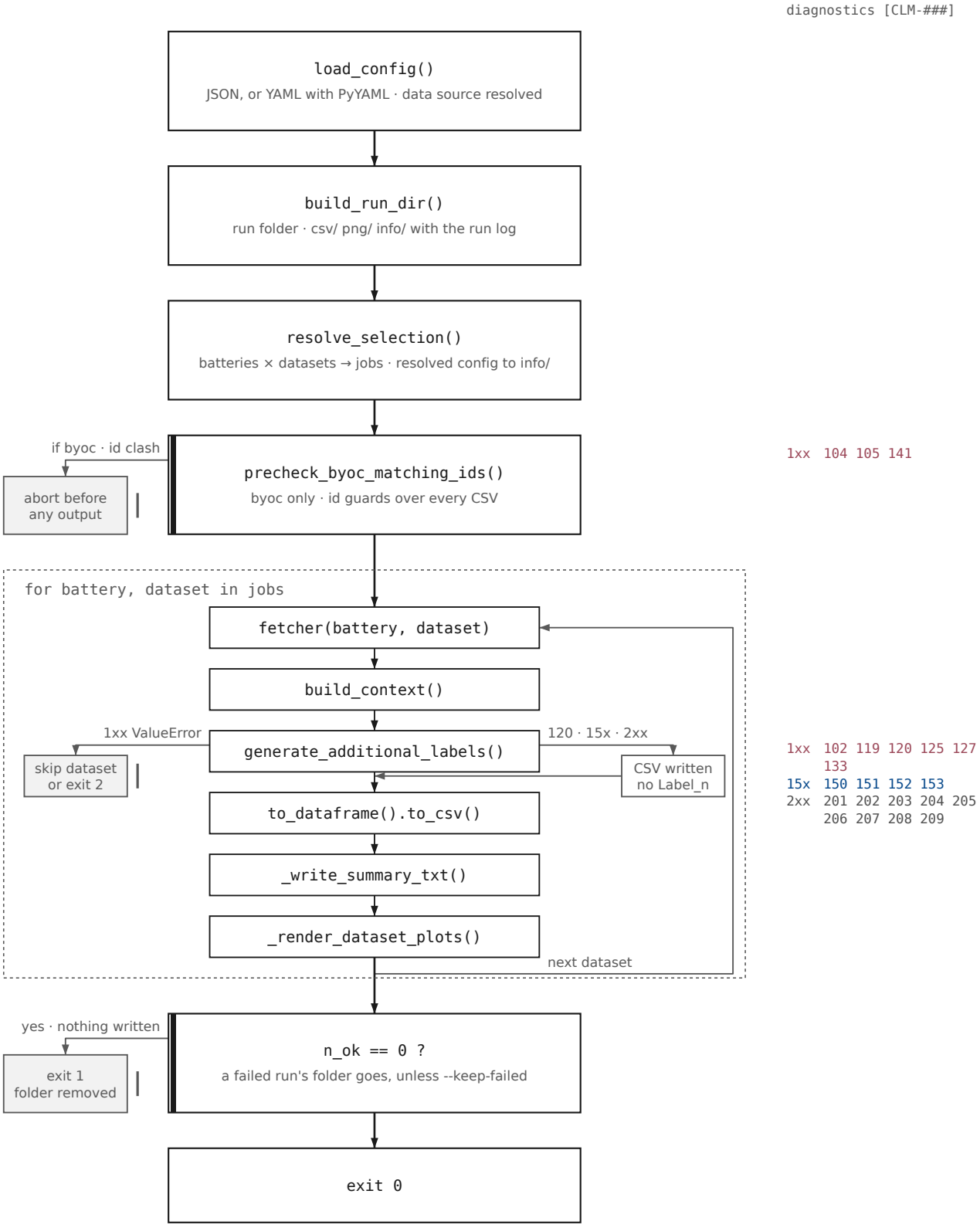
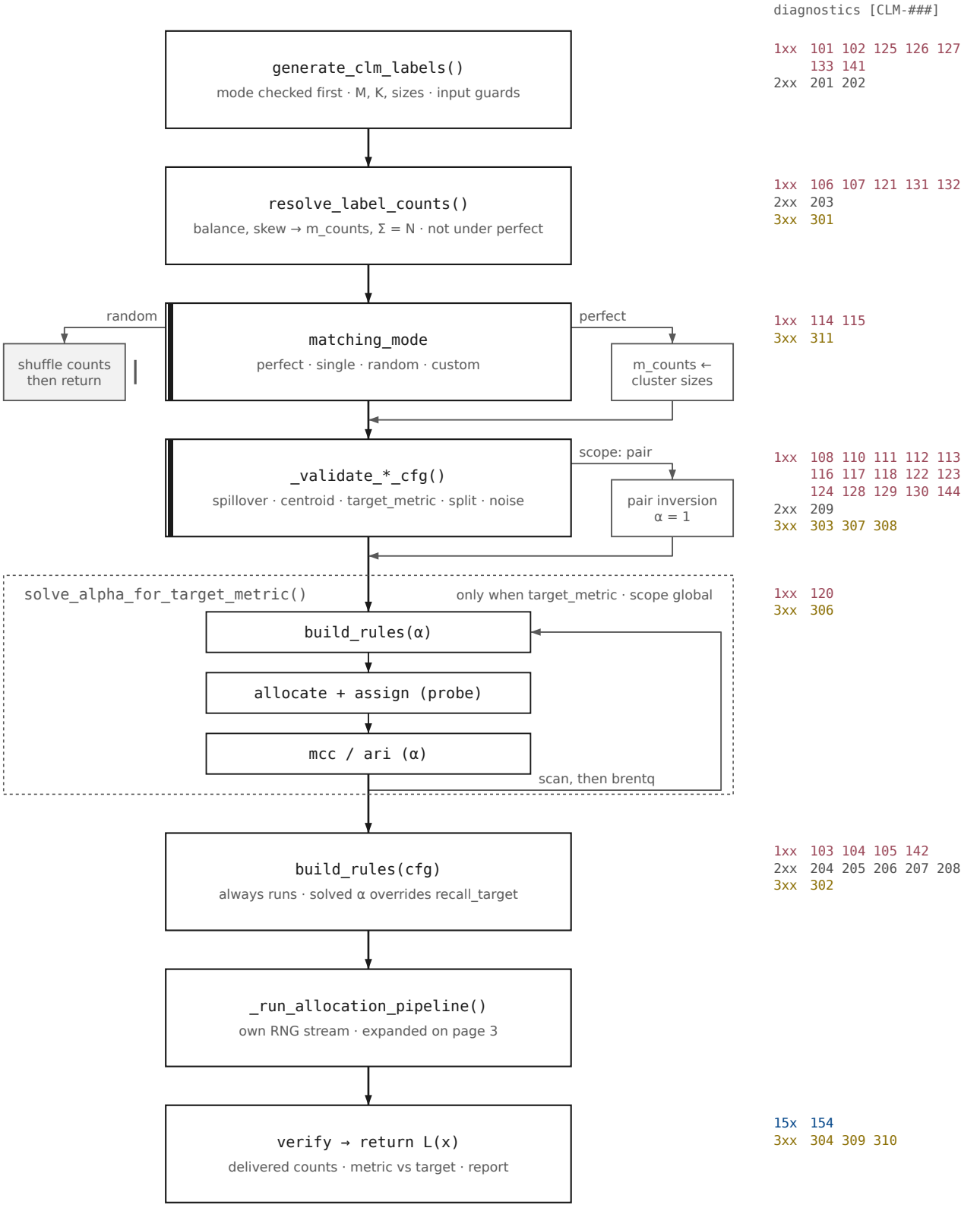


Figure A.1 · outer pipeline (1 of 3)

1xx invalid config · 15x infeasible allocation · 2xx missing key · 3xx warning

CLMSynth · clm_label_engine

c(x), X, cfg, seed → L(x) ∈ {0 … M−1}, deterministic under seed



diagnostics [CLM-###]

generate_clm_labels()	1xx 101 102 125 126 127 133 141 2xx 201 202
resolve_label_counts()	1xx 106 107 121 131 132 2xx 203 3xx 301
matching_mode	1xx 114 115 3xx 311
_validate_*_cfg()	1xx 108 110 111 112 113 116 117 118 122 123 124 128 129 130 144 2xx 209 3xx 303 307 308
solve_alpha_for_target_metric()	1xx 120 3xx 306
build_rules(cfg)	1xx 103 104 105 142 2xx 204 205 206 207 208 3xx 302
verify → return L(x)	15x 154 3xx 304 309 310

continues · stops here · branches

run or dataset stops · returns to the spine · one step, expanded

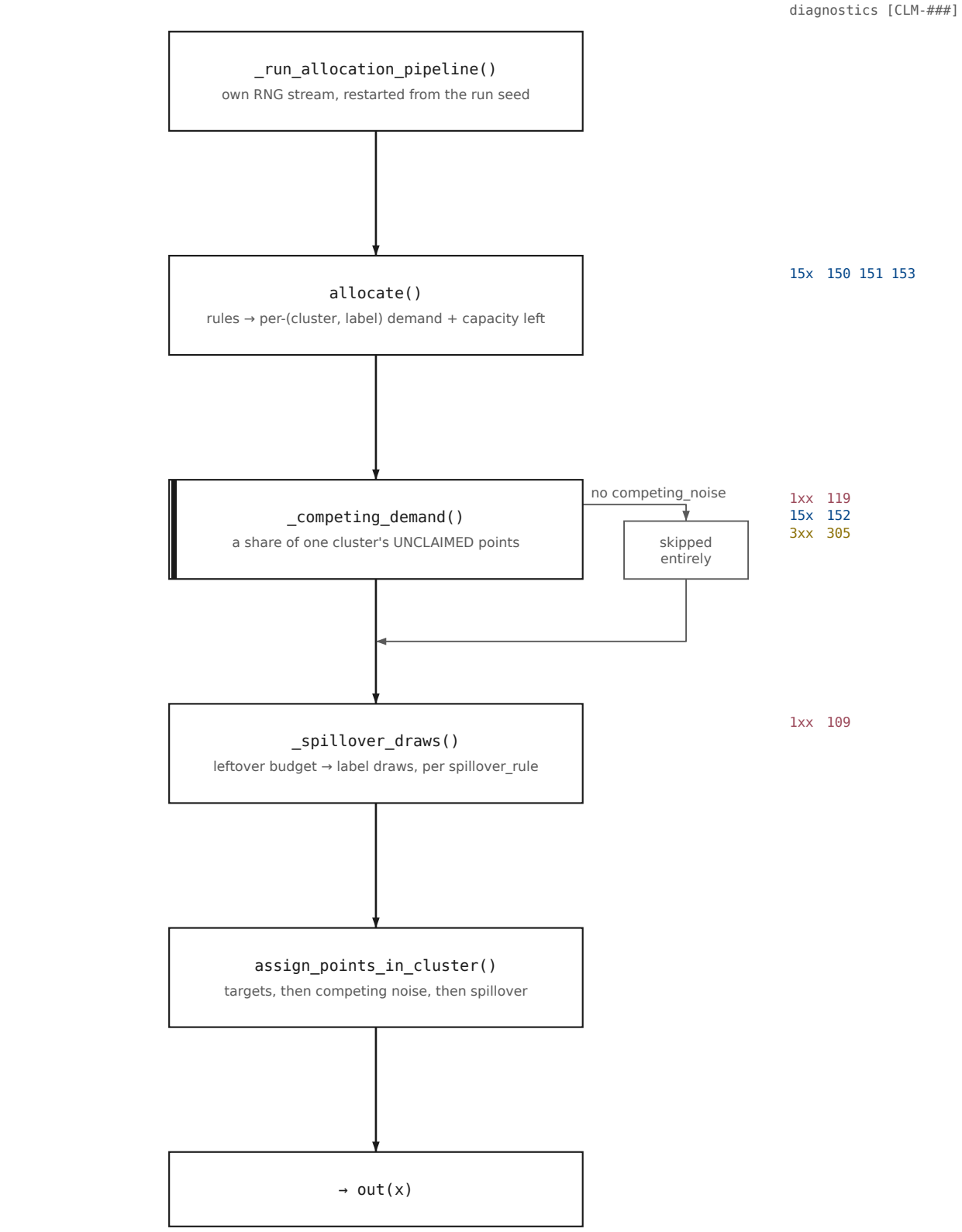
Figure A.2 · engine internals (2 of 3)

1xx invalid config · 15x infeasible allocation · 2xx missing key · 3xx warning

CLMSynth · _run_allocation_pipeline

rules, m_counts, cluster_sizes → out(x), one label per row

CLMSynth v0.7.5



continues · stops here · branches

run or dataset stops · returns to the spine · one step, expanded

Figure A.3 · allocation and placement (3 of 3)

1xx invalid config · 15x infeasible allocation · 2xx missing key · 3xx warning